

HW11_DSA8420

April 14, 2025

Primary Task: Set up a (mixed) integer linear programming model for each of the following problems. Make sure you state the definition of the variables. Do not solve the problem.

```
[1]: # libs
import numpy as np
from scipy.optimize import linprog
```

- 1 (10 points) A chair manufacturer makes three different types of chairs, each of which must go through sanding, staining, and varnishing. In addition, the model with the vinyl-covered back and seat must go through an upholstering process. Table 1 gives the time required for each operation on each type of chair, the available time for each operation in hours per month, and the profit per chair for each model. How many chairs of each type should be made to maximize the total profit?

Table 1

Model	Sanding(hrs)	Staining(hrs)	Varnishing(hrs)	Upholstering(hrs)	Profit(\$)
A-solid back & seat	1.0	0.5	0.7	0	10
B-ladder back, solid seat	1.2	0.5	0.7	0	13
C-Vinyl back & seat	0.7	0.3	0.3	0.7	8
Net available time per month	600	300	300	140	

Answer:

We will define our decision variables are as follows...

x_A : # of Model A chairs to make.

x_B : # of Model B chairs to make

x_C : # of Model C chairs to make

We can define our objective function to maximize total profit using our decision variables as...

$$\text{Max}Z = 10x_A + 13x_B + 8x_C$$

Which will be subject to the following constraints:

Sanding Time

$$1.0x_A + 1.2x_B + 0.7x_C \leq 600$$

Staining Time

$$0.5x_A + 0.5x_B + 0.3x_C \leq 300$$

Varnishing Time

$$0.7x_A + 0.7x_B + 0.3x_C \leq 300$$

Upholstering Time

$$0.7x_C \leq 140$$

```
[2]: # Problem 1 setup

# max 10x_A + 13x_B + 8x_C ; flipping coeff since linprog does min instead of ↪max
c = -np.array([10, 13, 8])
A_ub = np.array([
    [1.0, 1.2, 0.7],
    [0.5, 0.5, 0.3],
    [0.7, 0.7, 0.3],
    [0.0, 0.0, 0.7]
])
b_ub = np.array([600, 300, 300, 140]) # RHS
bounds = [(0, None), (0, None), (0, None)]
```

- 2 (10 points) A company is considering several investment opportunities, each of which differs in the initial capital required and cannot be invested repeatedly. The accounting department has done a thorough analysis of each of these investments and has estimated the long-term profit of each. The initial capital required and estimated profits (in millions of dollars) are summarized in Table 2. *Investments 1 and 4 are considered high-risk investments and management has decided to invest in at most one of these.* In addition, *Investment 6 is contingent upon also investing in Investment 3*, that is, Investment 6 is invested only if Investment 3 is invested. If \$120 million of initial capital is available, formulate an integer programming model to determine the optimal investment strategy.

Table 2

Investment	Initial Capital	Estimated Profit
1	26	18
2	34	12
3	18	7
4	45	24
5	31	11
6	39	15
7	23	9
8	13	6

Answer:

Unlike our previous question, we are now working with binary decision variables for investment i where i is one of 8 investments. I will define our binary decision variables as follows.

Let $x_i \in 0, 1$ for investment i , where...

$x_i == 1$ if investment chosen. $x_i == 0$ if investment not chosen.

We can define our objective function to maximize net estimated profit using our binary decision variables as...

$$MaxZ = 18x_1 + 12x_2 + 7x_3 + 24x_4 + 11x_5 + 15x_6 + 9x_7 + 6x_8$$

Subject to constraints:

2.1 Capital constraint

$$26x_1 + 34x_2 + 18x_3 + 45x_4 + 31x_5 + 39x_6 + 23x_7 + 13x_8 \leq 120$$